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Unit 3 - Repetition Statements

PDF Documents

PDF Document 1: 1734183039273-Chapter 3_unit3 Repetition Statements.pdf

This PDF document is included in the unit materials.

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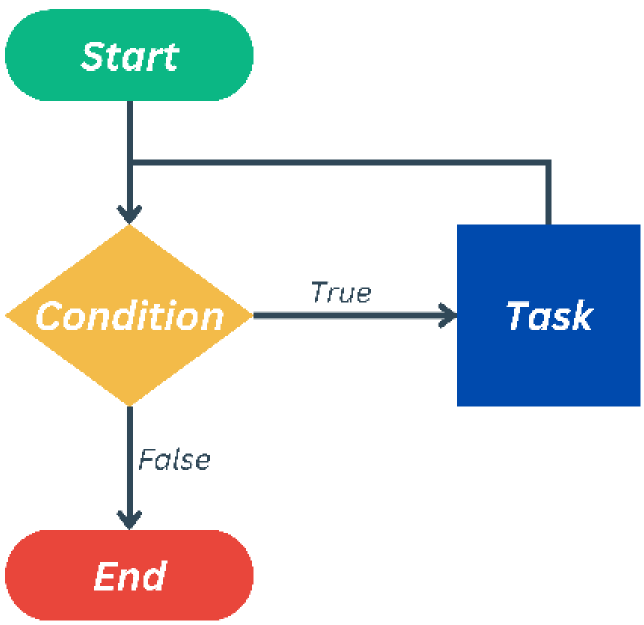
UNIT 3 REPETITION STATEMENTS

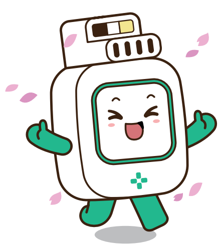
While statement

A "while" loop continues to execute its block of code as long as the condition is true. It checks the condition before running the loop's code. If the condition is true, the loop runs; if it's false, the loop stops. For example, "while the sun is up, keep the LED off."

✓ WHILE

Everyday Life Example





Imagine you have a playlist you love listening to while doing your homework. You decide that you will keep listening to music while you have homework left to do. So, here's what happens:

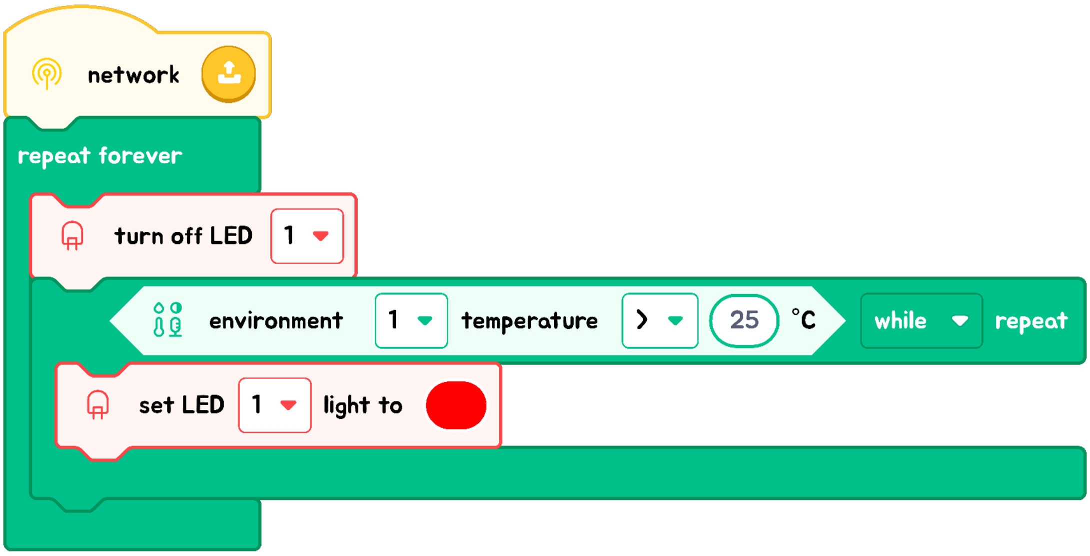
1. Check if there's homework left: This is like the condition in the "while" loop.

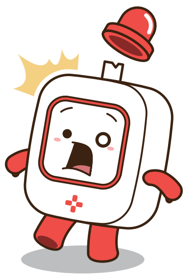
2. Listen to a song: If you still have homework, you play a song. This is like the code inside the loop.

3. Repeat: When the song ends, you check if there's still homework left. If there is, you play the next song.

4. Stop: Once all your homework is done, you stop playing music. The "while" loop ends because the condition (having homework) is no longer true.

EXPLANATION WITH CODE EXAMPLE





In the code, an "if" statement checks if the temperature exceeds 25 degrees Celsius. If true, it activates a red light as a warning. To maintain this warning as long as the temperature remains high, a "while" loop is used, ensuring the red light continues to signal the warning consistently.

Code Insight

It's important to distinguish between "if" statements and "while" loops:
an "if" statement evaluates a condition once and performs a task if that condition is true. A "while" loop, however, repeatedly executes a task as long as its condition stays true, continuously checking the condition and performing the task each cycle.

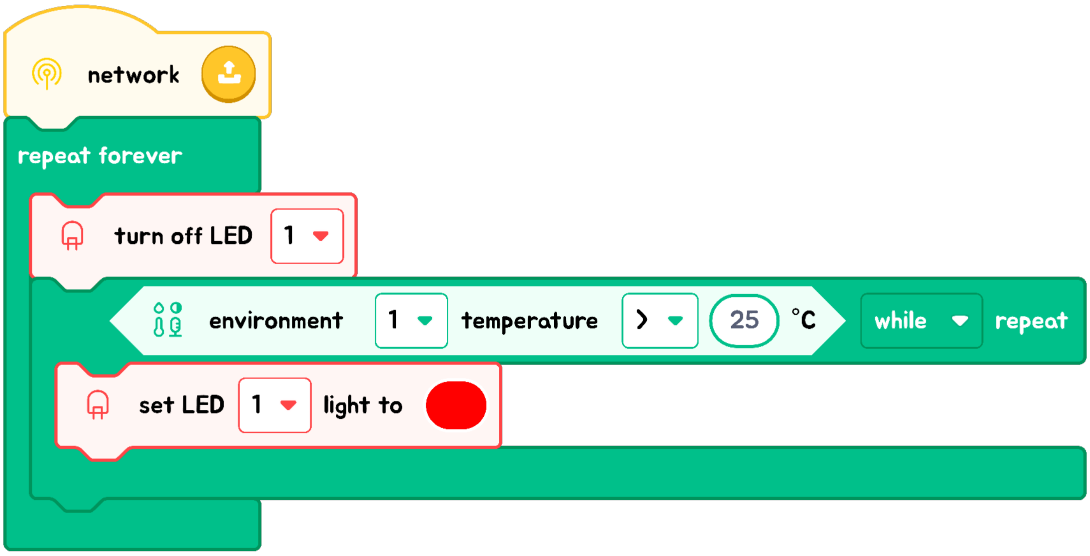
TIPS

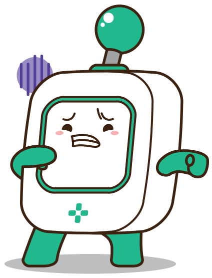
The temperature readings from the environmental module don't adjust as quickly as using a dial or button because external environmental changes aren't immediately responsive. It's not like getting instant feedback; rather, it involves waiting for changes to gradually become apparent.

Exercise

Flowchart

Draw a flowchart based on the code shared below:





YOUR FLOWCHART

Multiple-Choice Question

What does a while loop do in programming?

- (A) Executes a task once if a condition is true.
- (B) Repeatedly executes a task as long as a condition is true.
- (C) Executes a task only if multiple conditions are true.
- (D) Stops executing as soon as the condition becomes false.

Find Examples

Think of an example from daily life where a "while" loop could be used. Describe how it works similarly to a "while" loop in programming.